

OENG1250 Part-A Individual Design Journal — Powerpack & Solar

Student name: Hugh Williams

Student number: [REDACTED]

Group: 184

Rover component: Powerpack

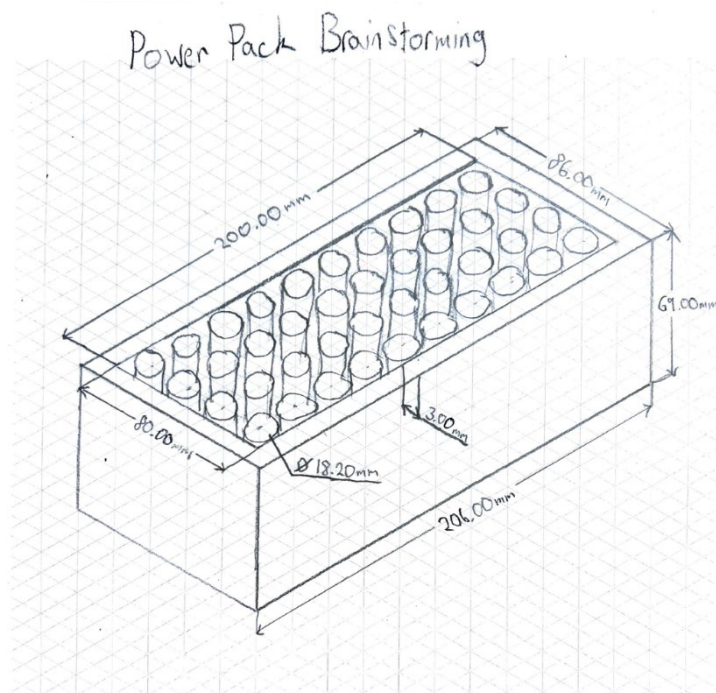
Problem statement

The aim of this project is to design a powerpack that can fulfill the Lunar Rover's energy storage requirements. It must have the power capacity to support the rover's operations whilst also being compatible in the assembly when attached to the body and other subsystems.

The powerpack is restricted to using cylindrical 18650 lithium-ion cells with the dimensions $\text{Ø}18.2\text{mm} \times 65.1\text{mm}$ which are to be held in the rover body. The solar panel used is assumed to be a constant 2.15 kg and the weight of the power pack assumes the mass density of lithium-ion is 2700 kg/m^3 .

Brainstorming and concept generation

Sketch gallery

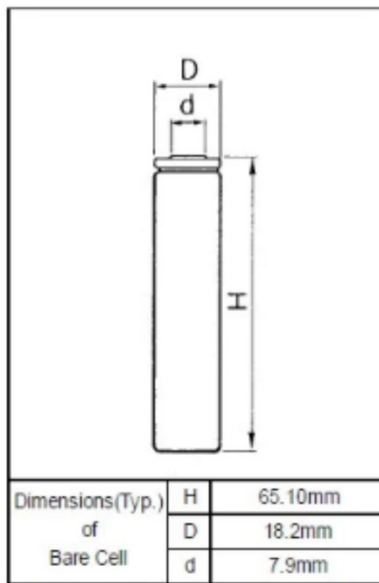


Early dimensions and connecting features

Dimensions can be referred to using the sketch above. For connecting the powerpack to the body of the rover, it is recommended to use m6 sized bolts.

Powerpack calculations and sizing summary

Calculations for the powerpack sizing can be found below. Additionally, for references below is the image of the dimensions of a bare cell.



$$V = 12 - \text{volt, motor consumes } 3 \text{ W at } 10 \text{ RPM}$$

$$5 \text{ Ah} \rightarrow 5 \text{ A for } 1 \text{ hour}$$

$$t = 12.5 \times S.F. = 12.5 \times 1.5 = 18.75$$

$$P = IV \rightarrow I = \frac{P}{V} = \frac{3}{12} = 0.25 \text{ A}$$

$$I_{Total} = 6 \times 0.25 = 1.5 \text{ A} \rightarrow 18.75 \text{ h}$$

$$IT = 1.5 \times 18.75 = 28.125 \text{ Ah} \rightarrow 30 \text{ Ah} \rightarrow 30000 \text{ mAh}$$

$$E^I = 250 \frac{Wh}{Kg}$$

$$E = TIV = 30 \times 12 = 360 \text{ Wh}$$

$$m = \frac{E}{E^I} = \frac{360}{250} = 1.44 \text{ kg}$$

$$E_V = 559 \text{ Wh/l}$$

$$V = \frac{E_V}{E_V^I} = \frac{360}{559} = 0.644 \text{ l}$$

$$1000 \text{ l} = 1 \text{ m}^3$$

$$1 \text{ l} = 10^{-3} \text{ m}^3$$

$$V = 0.644 \text{ l} \times 10^{-3} = 644 \times 10^{-6} \text{ m}^3$$

$$1 \text{ m}^3 = 10^9 \text{ mm}^3$$

$$V = 644 \times 10^{-6} \times 10^9 = 644 \times 10^3 \text{ mm}^3$$

$$V^I = \pi \left(\frac{d}{2} \right)^2 l = \pi \left(\frac{18.2}{2} \right)^2 \times 65 = 16901 \text{ mm}^3$$

$$N = \frac{V_T}{V^I} = \frac{644 \times 10^3}{16901} = 38.1 \approx 40$$

$$L = 65 \text{ mm}$$

$$W = 4 \times 18.2 \approx 80 \text{ mm}$$

$$L = 10 \times 18.2 \approx 190 \text{ mm}$$

This also gives 0.8 mm room between each cell when in the powerpack because it is unrealistic to assume they will be sized exactly and room for error must be allowed.

Weight Calculations

$$\rho = 2700 \text{ kg/m}^3$$

$$\text{Cell Volume} = 40 \times 1184.82 = 47392.80 \text{ mm}^3 \rightarrow 4.73928 \times 10^{-5} \text{ m}^3$$

$$\text{Cell Mass} = 2700 \times (4.73928 \times 10^{-5}) = 0.12796 \text{ kg} \rightarrow 127.96 \text{ g}$$

$$\text{Case Mass} = 396.54 \text{ g}$$

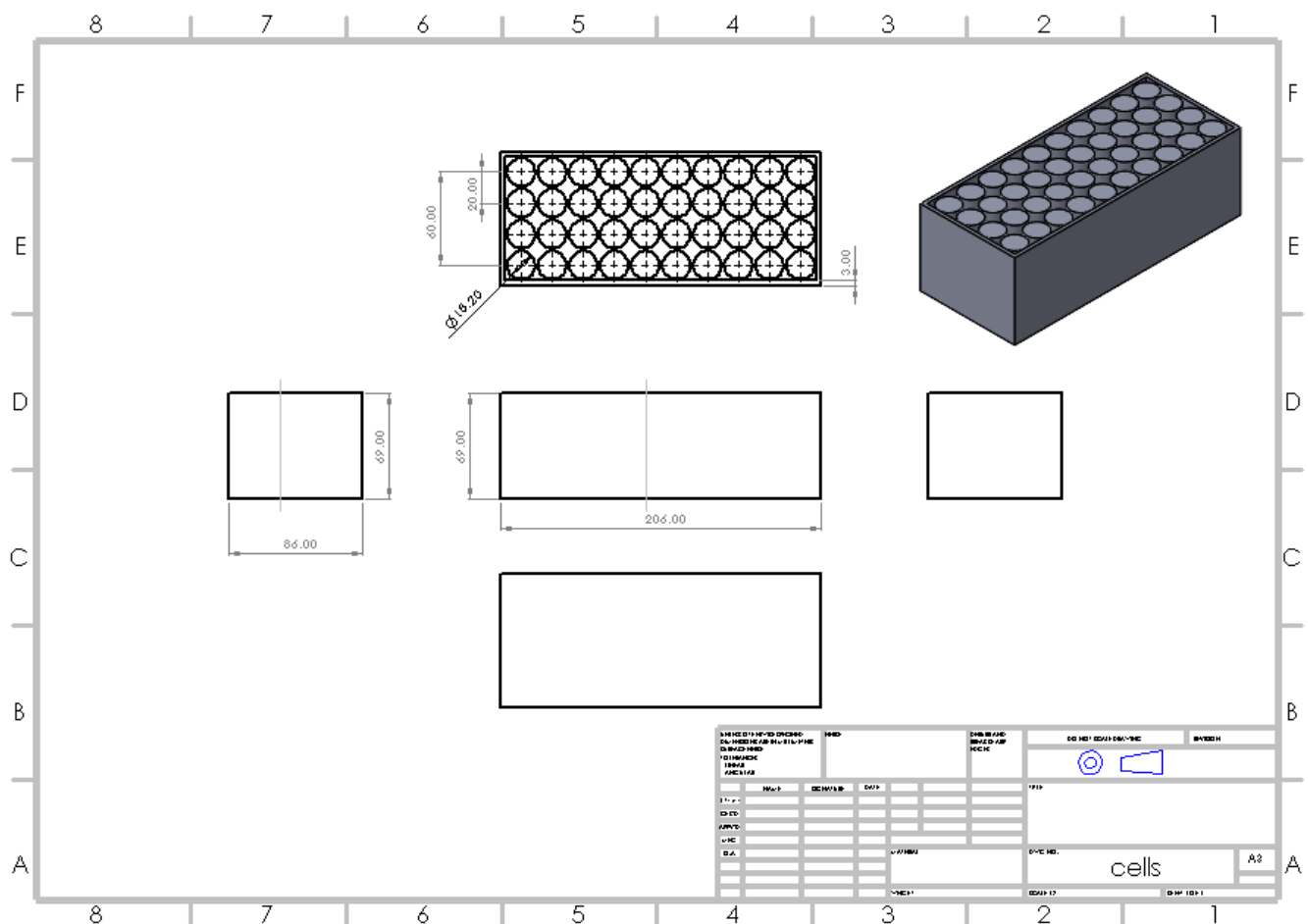
$$\text{Total Mass} = 127.96 + 396.54 = 524.50 \text{ g}$$

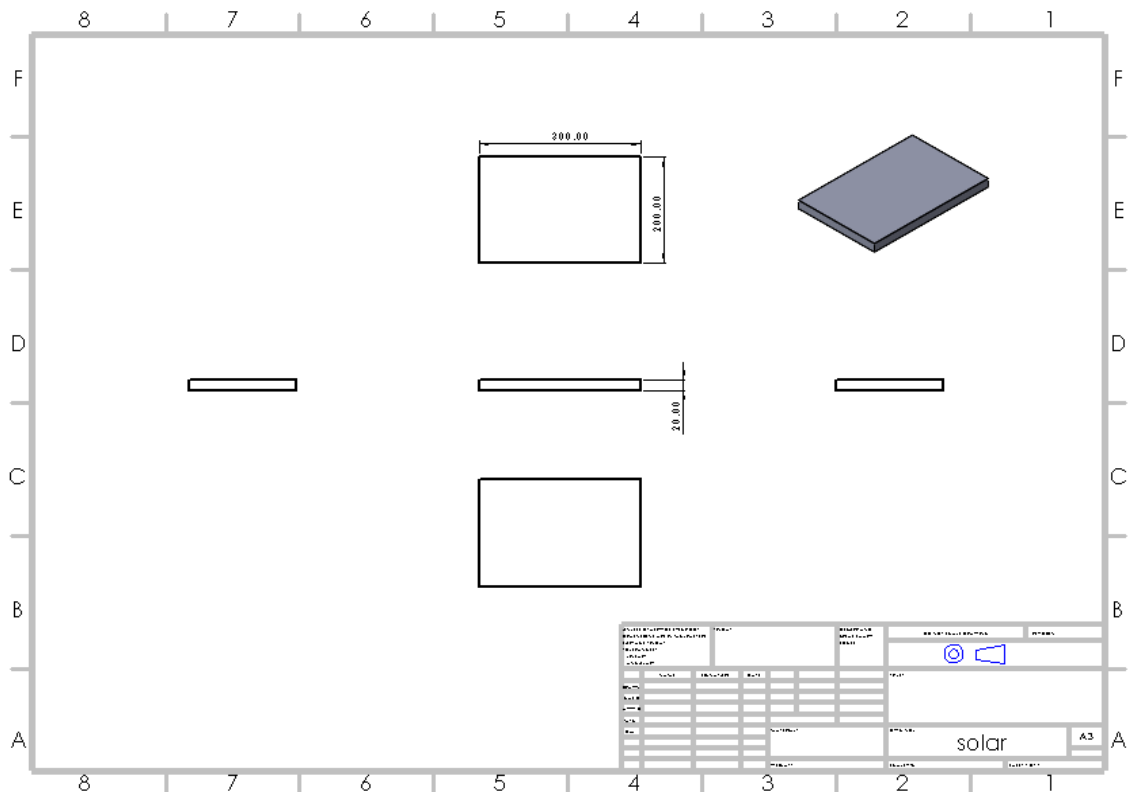
Rover Sub-assembly Model

Assembly mates and constraints

The sub-assembly uses 8 concentric mates to align the body and the connector parts of the powerpack with the M6 bolt joining them. There is also a coincident mate to make the face on the body and the base of the powerpack coplanar.

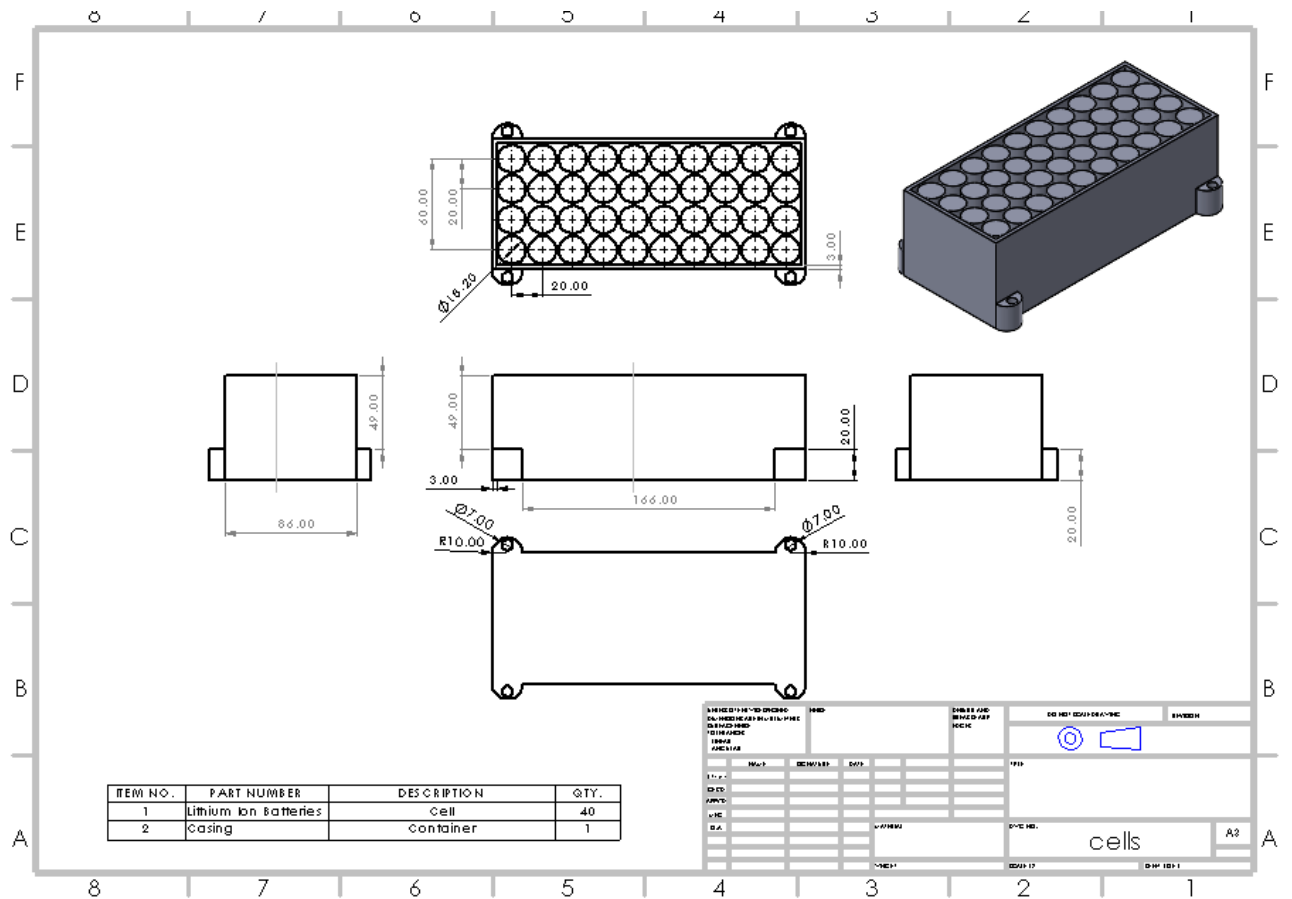
Preliminary design drawing



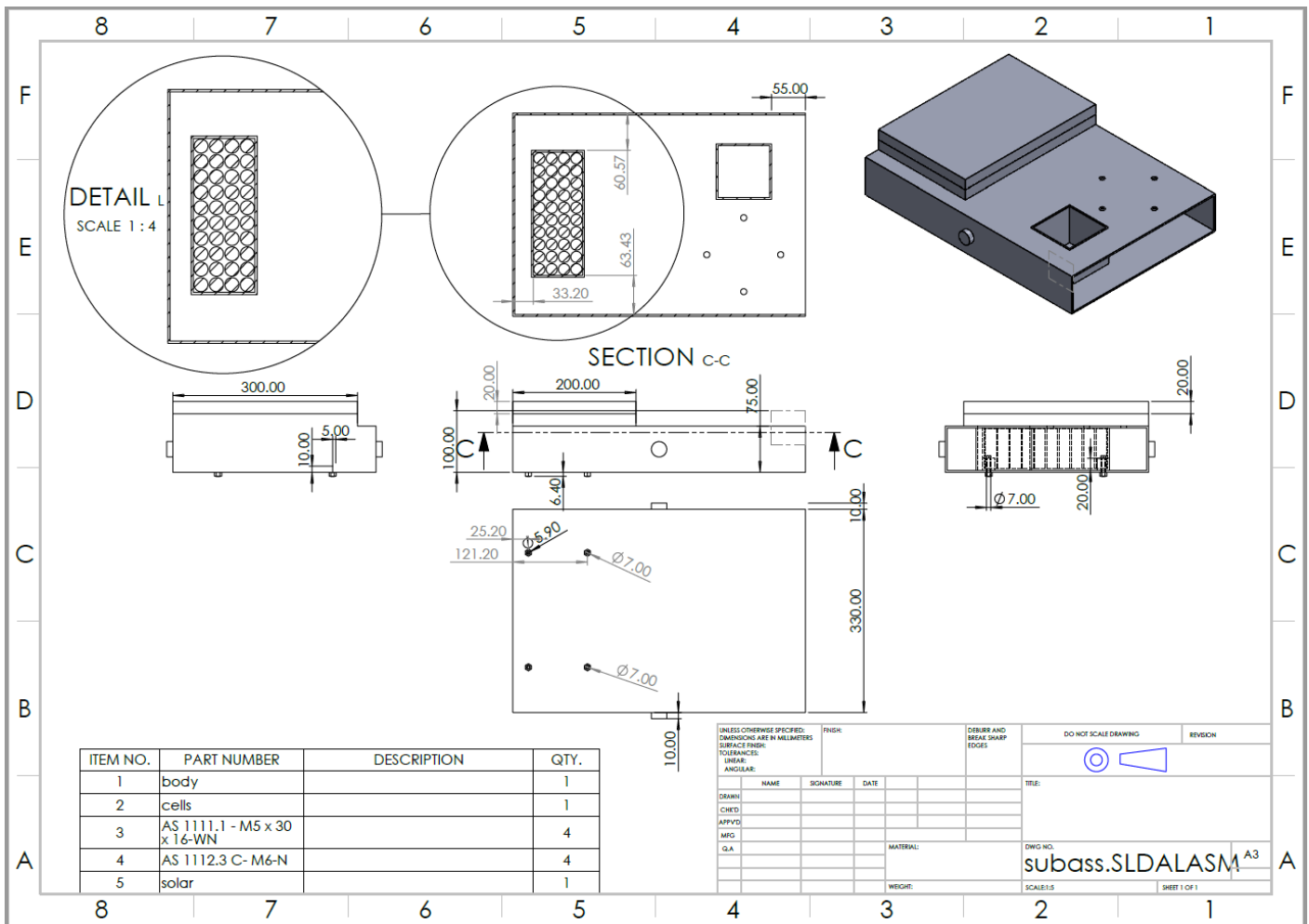


Solar

Detailed design drawing



Rover sub-assembly drawing



Meeting minutes, Week 3-8

Group 184					
Name	Student No	Student Signature	Contribution		
			Part-A Preliminary Model Deadline	Part-A Detailed Model Deadline	No of Meetings Addended (Wk 3 to 8)
		Kayle	Met	Met	5
			Met	Met	5

		Stirling	Met	Met	5
		Hugo	Met	Met	5
		Lucas	Met	Met	5
Hugh Williams		Hugh	Met	Met	5